

PHASE I: Project Title:

Kali Linux Digital Forensics Lab for Investigating a Suspicious ZIP File

1. Introduction

In this project, I will be setting up a digital forensics environment using Kali Linux to gain hands-on experience with forensic tools and investigation techniques. The goal is to understand how digital evidence is collected, examined, and documented in a structured way.

For this investigation, I will simulate a real-world scenario involving a suspicious ZIP file that was found on a system without a clear origin. The contents of the file will be analyzed to determine whether it contains hidden data or sensitive information. This project will help me build practical skills that can be applied in cybersecurity and forensic analysis.

2. Kali Linux Installation Plan

2.1 Installation Method

For this project, I will use Kali Linux installed on a virtual machine (VirtualBox) running on my personal laptop. This allows me to work in a safe environment without affecting my main operating system.

2.2 Justification

I chose to use a virtual machine instead of a cloud service because it gives me full control over the system and makes it easier to practice locally. It also allows me to take snapshots, which is useful if I need to restore the system to a previous state. This setup works best for my current workflow and makes testing forensic tools more convenient.

2.3 Pre-installation Steps

Before installing Kali Linux, I will complete the following steps:

- Verify that my laptop meets system requirements (RAM, storage, CPU)
- Download and install Oracle VirtualBox
- Download the official Kali Linux ISO file
- Create a new virtual machine in VirtualBox
- Allocate system resources (RAM, CPU, and storage)
- Attach the Kali Linux ISO file to begin installation

2.4 Post-installation Configuration

After installation, I will:

- Update the system using apt update and apt upgrade
- Configure user accounts and system settings
- Verify network connectivity
- Ensure forensic tools are available or install any missing tools
- Take a snapshot of the clean system for backup

3. Forensic Tools Identification

3.1 Disk Imaging Tools

- **FTK Imager**

Used to create exact copies of storage devices while preserving data integrity for analysis.

- **dc3dd**

A command-line imaging tool that supports hashing and logging to ensure accurate data acquisition.

- **Autopsy**

A forensic tool used to analyze disk images, recover deleted files, and examine system activity.

3.2 Live Acquisition Tools

- **dd**

A basic command-line tool used to copy and acquire data from storage devices.

- **LiME (Linux Memory Extractor)**

Used to capture memory from a running system for further analysis.

3.3 Memory Analysis Tools

- **Volatility**

A widely used tool for analyzing memory dumps to identify processes, connections, and suspicious activity.

- **Rekall**

A memory forensic tool used for fast analysis during incident response.

3.4 Network Analysis Tools

- **Wireshark**

Used to capture and analyze network traffic in real time.

- **tcpdump**

A command-line tool used for capturing network packets.

3.5 Malware Analysis Tools

- **Ghidra**

A reverse engineering tool used to analyze suspicious or malicious programs.

- **Radare2**

A framework used for analyzing binary files and performing reverse engineering tasks.

3.6 Steganography Tools

- **Steghide**

Used to detect and extract hidden data inside files.

- **zsteg**

A tool used to identify hidden data within image files.

3.7 Rationale

These tools were selected because they are commonly used in digital forensics and are available in Kali Linux. They cover multiple areas of investigation, including disk analysis, memory analysis, network monitoring, and hidden data detection. This combination of tools will allow me to perform a complete forensic investigation from start to finish.

4. Evidence Acquisition and Management Strategy

4.1 High-Level Overview

The investigation will focus on analyzing a suspicious ZIP file that contains multiple files. The process will involve extracting the archive, identifying all files (including hidden ones), and examining their contents for any sensitive or unusual data.

4.2 Best Practices

During this investigation, I will follow key forensic best practices, including:

- Maintaining proper documentation of all steps taken
- Avoiding unnecessary changes to the evidence
- Identifying hidden files and data
- Reviewing all file contents carefully
- Organizing findings clearly for reporting

5. Conclusion

This proposal outlines my plan to create a Kali Linux-based forensic lab and investigate a suspicious ZIP file. By setting up a controlled environment, selecting appropriate tools, and following structured procedures, I will be able to perform a realistic forensic investigation.

This phase prepares me to move into the next steps of the project, where I will apply these tools to acquire and analyze evidence in a practical scenario.